Policy Memo: Why Limiting Climate Change Matters for Niger

To: Minister of Environment, Head of Niger's UNFCCC delegation	**From:** KOUAME Koffi Fidèle, Climate Adviser
Date: 14 October 2026	**Subject:** Climate risks for Niger and the case for limiting global warming

Key message

Models indicate that dangerous heat in Niger is projected to rise from about 31 days a year today to 54 by 2050 under Current policies, against 45 under Below 2 °C and 41 under Net Zero 2050, the scenario consistent with the 1.5 °C goal. Extreme rainfall is also projected to intensify more under Current policies, but model ranges overlap, while severe drought is projected to affect almost the whole country in all three scenarios. Niger therefore has a direct interest both in faster global emission cuts and in adaptation finance now.

Three priority impacts

Impact (CIE indicator)	Today (2020)	2050: Net Zero 2050	2050: Below 2 °C	2050: Current policies
Dangerous heat: days per year with a daily maximum heat index above 40 °C	31 [25 to 38]	41 [29 to 60]	45 [33 to 69]	54 [41 to 84]
Severe drought: share of land with at least one month of SPEI below −1.5	84 % [72 to 97]	97 % [82 to 100]	99 % [85 to 100]	100 % [96 to 100]
Extreme rainfall: annual maximum 5-day rainfall, change from 1995–2014	+8 % [−7 to +29]	+16 % [−1 to +49]	+18 % [0 to +55]	+24 % [−1 to +67]

Table 1. Source: Climate Impact Explorer (Climate Analytics, 2026), Niger, annual values, area-weighted average. NGFS scenarios, with global warming in 2100 as given in the CIE data: Net Zero 2050 (1.3 °C), Below 2 °C (1.6 °C) and Current policies (2.9 °C). Multi-model median, with the 5–95 % range in brackets.

Dangerous heat. The indicator counts days on which the daily maximum heat index exceeds the "danger" threshold of 40 °C (Climate Analytics, 2026). By 2050, models project about 54 such days a year under Current policies, against 45 under Below 2 °C and 41 under Net Zero 2050, close to the 2030 level of about 39 days. Results after 2060 are only indicative, but the gap keeps widening: about 91 days [59 to 134] against 42 [29 to 72] by 2100. This matters because agriculture employs over 80 % of the population (World Bank, 2024), and farming and herding are outdoor work.

Severe drought. This indicator accounts for evaporation as well as rainfall. About 84 % of the land already experiences at least one month of severe drought a year; models project about 96 % by 2030 and 97 to 100 % by 2050 in all three scenarios, with largely overlapping ranges. Rain-fed millet, sorghum and pastures are therefore likely to face more frequent water stress whatever the global outcome.

Extreme rainfall. The heaviest five-day rainfall of the year could intensify by about 24 % by 2050 under Current policies, against 18 % under Below 2 °C and 16 % under Net Zero 2050. The ranges overlap and their low ends are close to zero, so this difference is uncertain. Heavier downpours would raise flood risk in Niamey and the Niger River valley, where people, infrastructure and irrigated rice are concentrated.

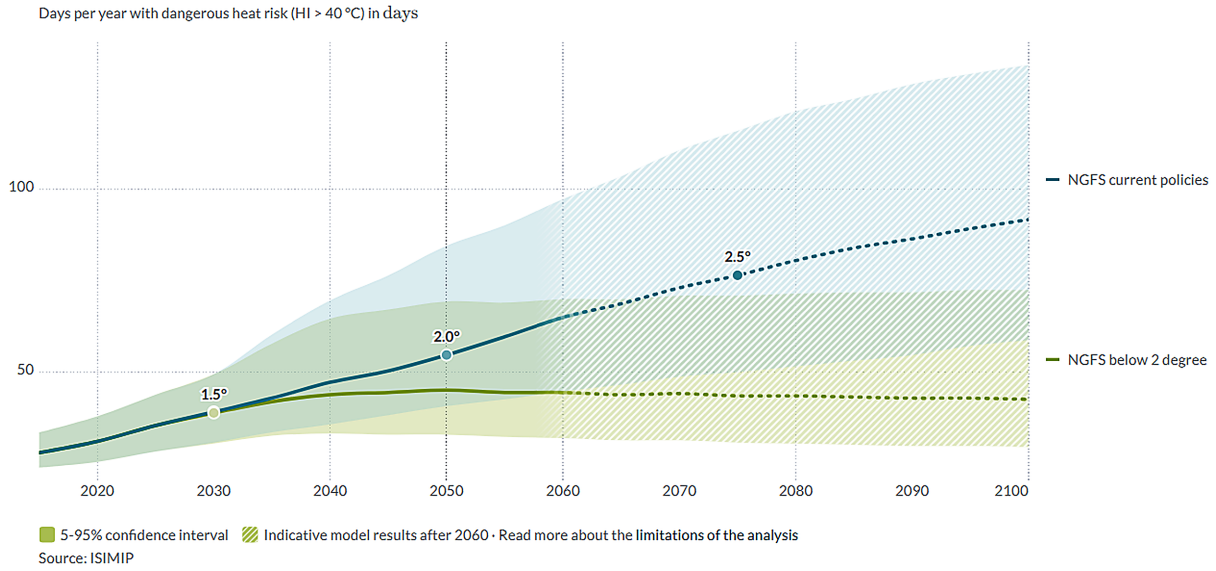


Figure 1. Days per year with dangerous heat risk in Niger, NGFS Current policies (blue) and Below 2 °C (green). Screenshot from the Climate Impact Explorer (Climate Analytics, 2026). The two curves are identical until about 2030 and then separate; the gap is what limiting warming avoids. Dots on the blue curve mark the years in which global warming reaches 1.5, 2.0 and 2.5 °C under Current policies. Shaded bands show the 5–95 % model range; after 2060 the lines are dotted and the bands hatched because results are indicative.

Implications for Niger's national interest

Of the three indicators, heat is the one on which global emission cuts make the clearest difference. By 2050, Below 2 °C would avoid about 9 days of dangerous heat a year compared with Current policies, and Net Zero 2050 about 13; after 2050, Net Zero 2050 is the only scenario in which heat days fall back towards today's level (about 32 by 2100, indicative). The World Bank estimates that climate change could reduce Niger's annual GDP by 2.2 % in wetter scenarios and by 11.9 % in drier ones by 2050. It links extreme weather to low agricultural production, high food prices and food insecurity, and notes growing competition between farmers and pastoralists for scarce resources (World Bank, 2022). Niger should therefore defend the 1.5 °C goal in the negotiations and, because drought is projected to spread in every scenario, seek adaptation finance for drought-tolerant crops, water harvesting and early warning.

Uncertainty

These are projections, not forecasts, with wide model ranges; this memo relies mainly on 2050 because the tool marks results after 2060 as indicative. The heat index relies on estimated humidity, and the drought indicator saturates. Given Niger's high vulnerability, actual impacts may exceed what these hazard indicators suggest.

References

Climate Analytics (2026). *Climate Impact Explorer*: Niger. Indicators: days per year with dangerous heat risk (HI > 40 °C), area under severe drought (SPEI < −1.5) and annual maximum 5-day precipitation; NGFS Net Zero 2050, Below 2 °C and Current Policies scenarios; ISIMIP data. https://climate-impact-explorer.climateanalytics.org (accessed 9 October 2026).

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